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(54) **MULTI-COMPONENT DEVICE AND
METHOD OF MAKING A
MULTI-COMPONENT DEVICE**

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(51) **Int. Cl.**

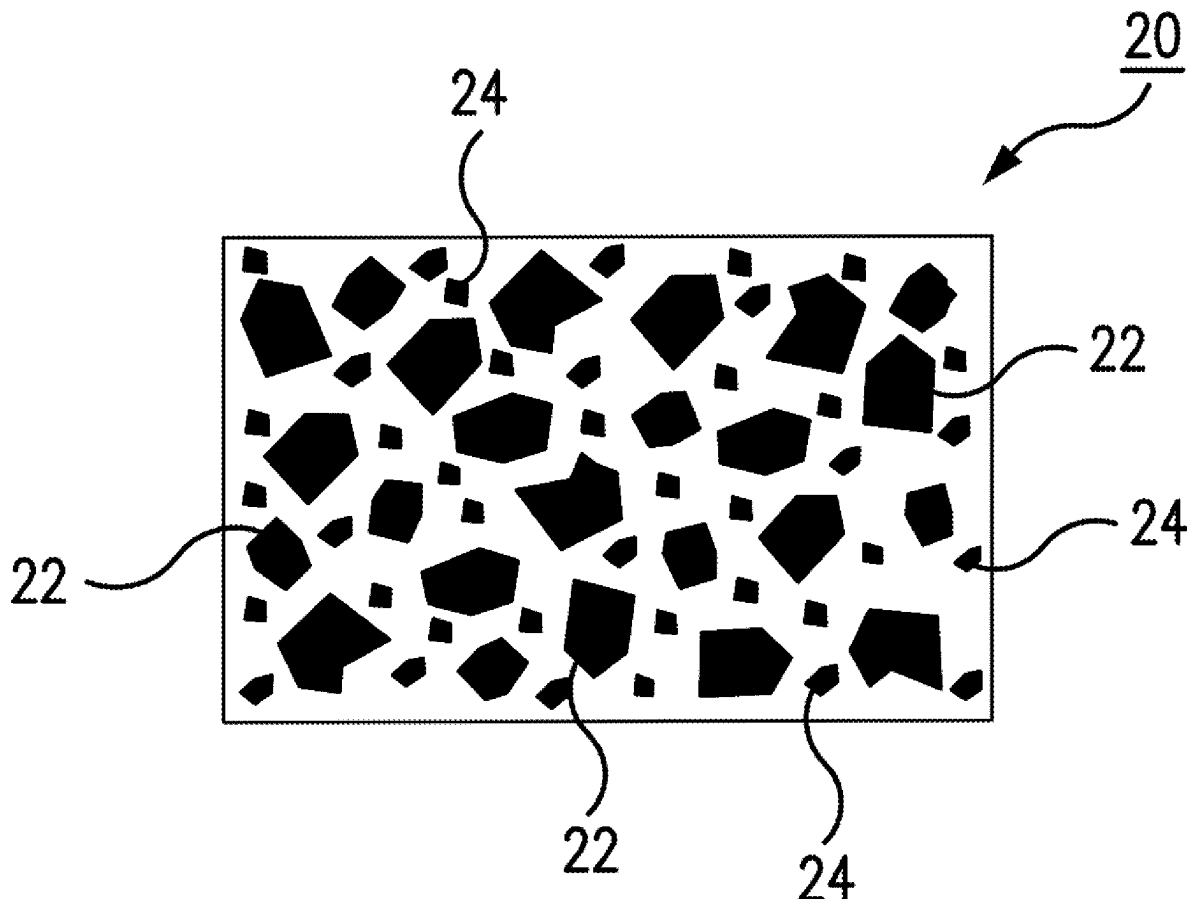
C04B 41/45 (2006.01)

C04B 41/50 (2006.01)

(57)

ABSTRACT

A device includes a ceramic substrate formed of a first material, a polishable layer formed of a different material, and an interface between the ceramic substrate and the polishable layer. The interface is formed by infiltration of molten elemental silicon, and bonds the ceramic substrate and the polishable layer together. The device may include an optical device such as, for example, mirror or a beam dump. A method of making a device from a green-state structure and a polishable layer is also disclosed. The method includes infiltrating elemental silicon into and through the green-state structure, to form a substrate of a multi-phase ceramic material from the green-state structure, and to reactively bond the substrate and the polishable layer together.



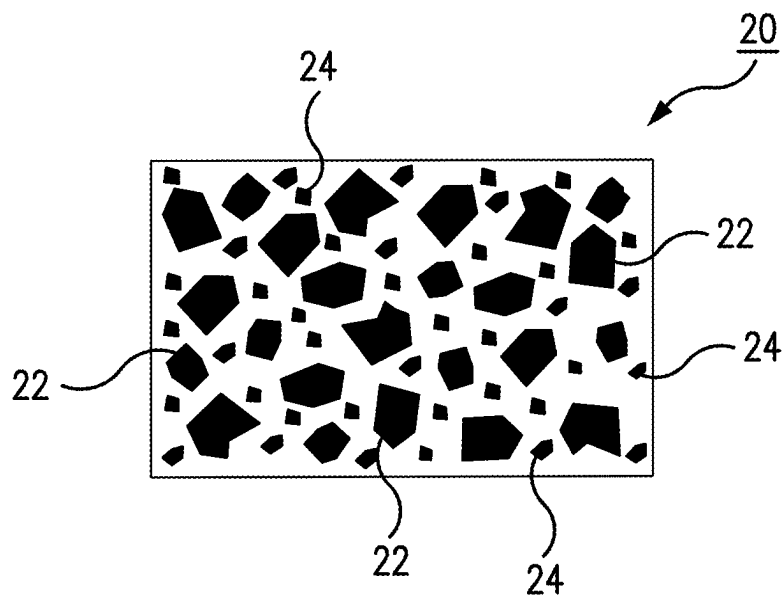


FIG. 1

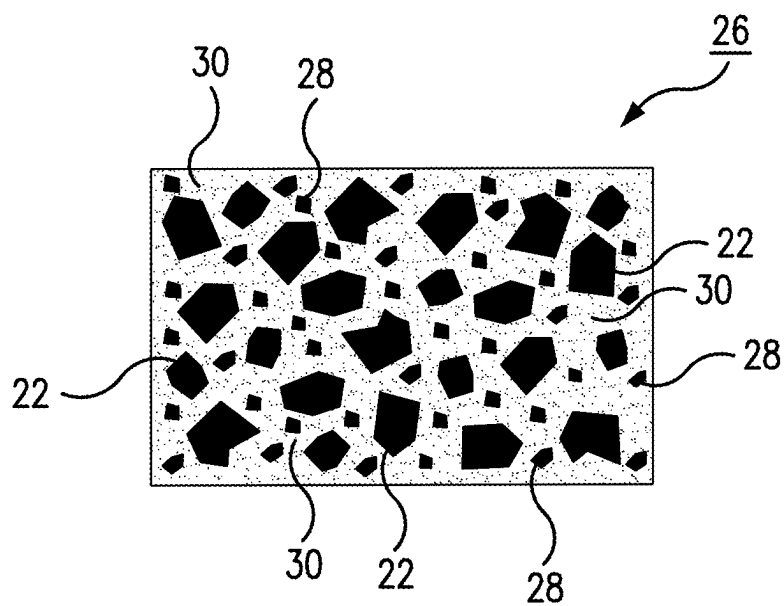


FIG. 2

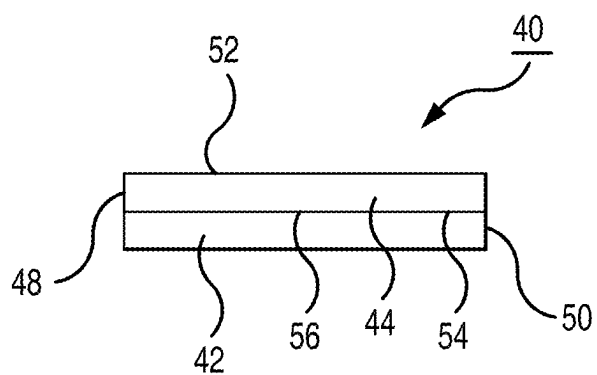


FIG. 3

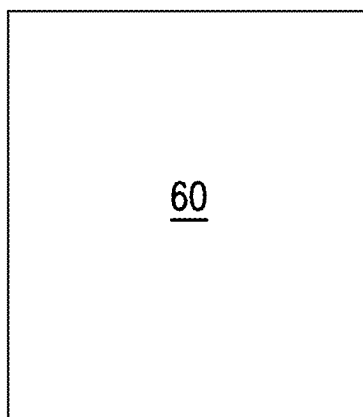


FIG. 4

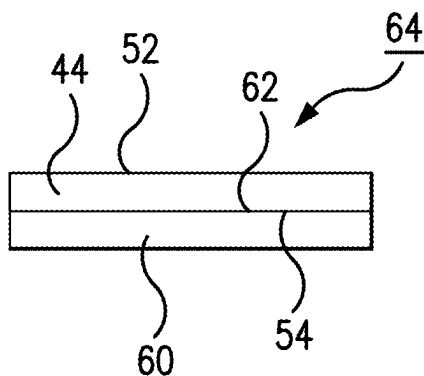


FIG. 5

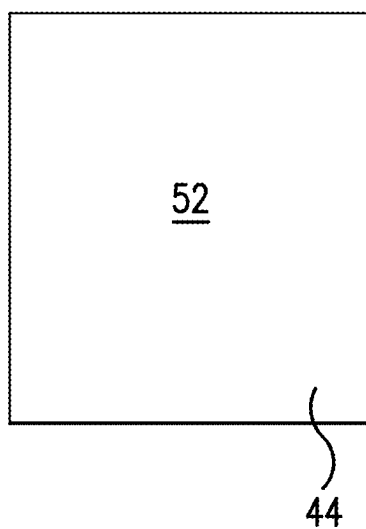


FIG. 6

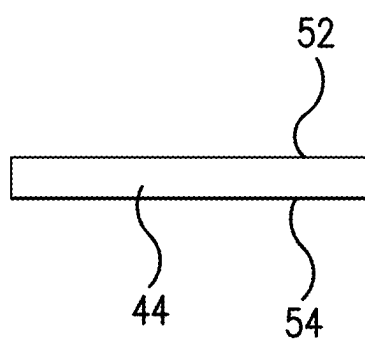


FIG. 7

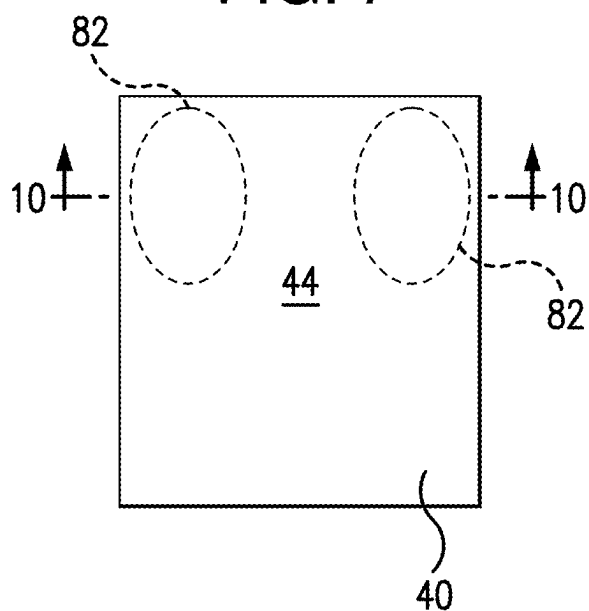


FIG. 8

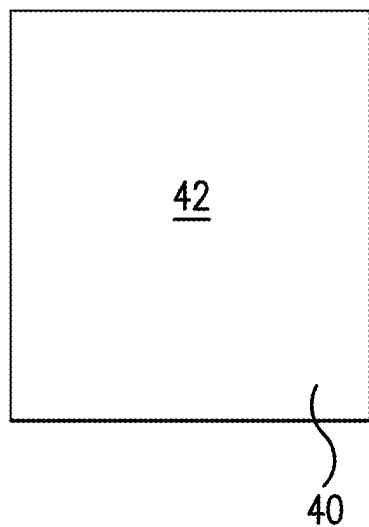


FIG. 9

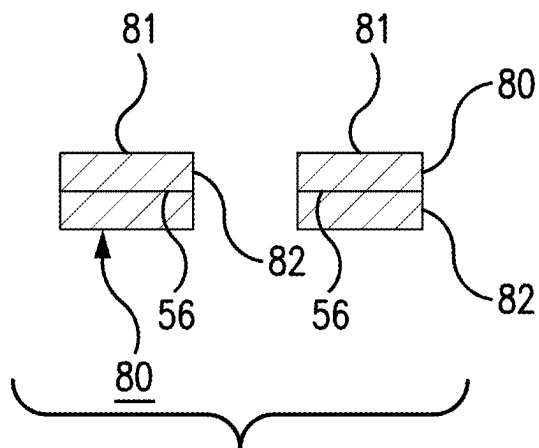


FIG. 10

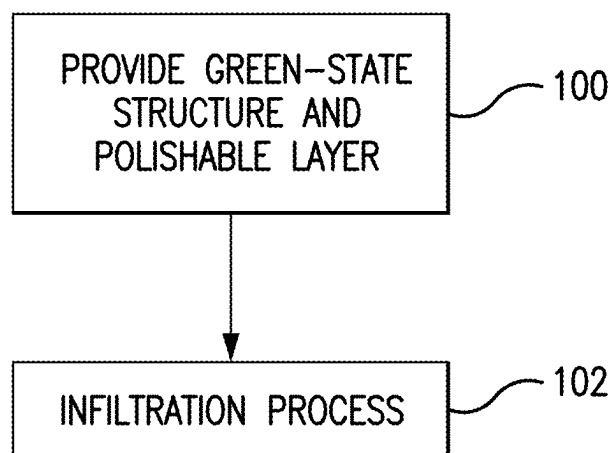


FIG. 11

MULTI-COMPONENT DEVICE AND METHOD OF MAKING A MULTI-COMPONENT DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Divisional of U.S. patent application Ser. No. 18/155,546, filed Jan. 17, 2023. The aforementioned application is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] Reaction-bonded silicon-carbide (RB-SiC, or Si/SiC) is a multi-phase material formed by reactive infiltration in which molten elemental silicon (Si) is brought into contact with a porous mass of interconnected silicon-carbide (SiC) particles plus carbon (C) in a vacuum or inert atmosphere. A wetting condition is created such that the molten silicon is pulled by capillary action into the mass of interconnected silicon-carbide particles and carbon, and the silicon reacts with the carbon in the mass to form additional silicon carbide. The resulting RB-SiC material contains primarily silicon carbide, but also unreacted, interconnected silicon.

[0003] The infiltration process is illustrated in FIGS. 1 and 2. FIG. 1 is a schematic cross-sectional view of the porous mass 20 which contains interconnected silicon-carbide particles 22 and carbon 24. FIG. 2 is a schematic cross-sectional view of the RB-SiC material (a ceramic material) 26 after infiltration by molten silicon, and subsequent cooling. The infiltration may occur in a vacuum or inert atmosphere.

[0004] In the illustrated process, molten silicon reactively infiltrates into the porous mass 20 (FIG. 1) such that the RB-SiC material 26 (FIG. 2) has a microstructure of three elements: (1) the original silicon-carbide particles 22; (2) reaction-formed silicon-carbide ($\text{Si} + \text{C} \Rightarrow \text{SiC}$) 28, and (3) residual (unreacted) elemental silicon 30. The silicon 30 expands as it solidifies from its liquid state such that the reacted, solidified three-component microstructure 22, 28, 30 is fully dense.

[0005] Thus, as the term is used herein, RB-SiC material is a fully dense, two-phase composite of silicon-carbide particles in a continuous silicon matrix.

[0006] RB-SiC material is mentioned in U.S. patent application Ser. No. 17/663,956 (filed May 18, 2022) and United States Patent Publications Nos. 2022/0227676 (published Jul. 21, 2022) and 2021/0331985 (published Oct. 28, 2021). The entire disclosures of U.S. patent application Ser. No. 17/663,956 and United States Patent Publications Nos. 2022/0227676 and 2021/0331985 are incorporated herein, in their entireties, by reference.

SUMMARY

[0007] The present disclosure relates to a multi-component device formed of a ceramic substrate bonded to a polishable layer, where the ceramic substrate and the polishable layer are formed of different materials. According to this aspect of the present disclosure, an interface is located between the ceramic substrate and the polishable layer, the interface bonds the ceramic substrate and the polishable layer together, and the interface is formed by infiltrating molten elemental silicon into and through a preform for the ceramic substrate.

[0008] According to another aspect of the present disclosure, the ceramic substrate is formed of a multi-phase material such as reaction-bonded silicon-carbide (RB-SiC). According to yet another aspect of the present disclosure, the polishable layer includes a polishable ceramic material such as chemical-vapor-deposited silicon-carbide (CVD-SiC) or single-crystal silicon-carbide (Xtal-SiC). The materials of the substrate and the polishable layer may be selected so that the substrate and the polishable layer have well-matched coefficients of thermal expansion (CTEs) across a wide range of temperatures. The substrate and the polishable layer may be considered to have well-matched CTEs when the difference in CTE between the substrate and the polishable layer is, for example, less than 2.0 ppm/ $^{\circ}\text{C}$., or less than 0.2 ppm/ $^{\circ}\text{C}$.

[0009] The present disclosure also relates to an optical device formed from a multi-component device, where the multi-component device includes a ceramic substrate bonded to a polishable layer, and the ceramic substrate and the polishable layer are formed of different materials. According to this aspect of the present disclosure, an interface is located between the substrate and the polishable layer. The interface bonds the substrate and the polishable layer together, and is formed by infiltrating elemental silicon, and the polishable layer has a polishable surface, facing away from the substrate. The polishable layer may be polished to reflect light.

[0010] The present disclosure also relates to a multi-component device which includes a light-absorbing beam dump. According to this aspect of the disclosure, the multi-component device includes a ceramic substrate bonded to a polishable layer, the ceramic substrate and the polishable layer are formed of different materials, and an interface is located between the substrate and the polishable layer. The interface bonds the substrate and the polishable layer together, and is formed by infiltrating elemental silicon, and the polishable surface faces away from the substrate.

[0011] The present disclosure also relates to a method of making a multi-component device, which includes the steps of: (1) providing a green-state structure and a polishable layer, wherein the green-state structure includes a porous mass of interconnected silicon-carbide and carbon; and (2) infiltrating elemental silicon into and through the green-state structure to form a substrate of a multi-phase ceramic material, and to bond the substrate and the polishable layer together. According to one aspect of the present disclosure, the substrate formed from the green-state structure includes a multi-phase material, and the polishable layer does not include reaction-bonded silicon-carbide.

[0012] If desired, the method of making the multi-component device includes the step of extracting a mirror from the bonded substrate and polishable layer. The extracting step may include wire electrical-discharge-machining (EDM).

[0013] The present disclosure also relates to a method of making a multi-component device. This method includes the steps of providing a porous mass of silicon carbide and carbon, causing molten elemental silicon to infiltrate the porous mass to form a multi-phase RB-SiC material, causing the molten silicon to flow through the porous mass and into contact with an unpolished surface of a polishable layer, and thereby bonding the polishable layer to the RB-SiC material. The polishable layer may be polished to a suitable finish

before or after the bonding process. The polishable layer may provide a secondary face material for the multi-component device.

[0014] The present disclosure also relates to a method of reaction bonding bulk CVD-SiC or Xtal-SiC to a RB-SiC substrate.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 is a schematic cross-sectional view of an example of a porous mass constructed in accordance with the present disclosure;

[0016] FIG. 2 is a schematic cross-sectional view of a reaction-bonded silicon-carbide (RB-SiC) material made from the porous mass of FIG. 1;

[0017] FIG. 3 is a side view of an example of a multi-component device constructed in accordance with the present disclosure;

[0018] FIG. 4 is a top view of an example of a green-state structure constructed in accordance with the present disclosure;

[0019] FIG. 5 is a side view of an example of a combined preform including the green-state structure of FIG. 4 and secondary face material, the latter being an example of a polishable layer constructed in accordance with the present disclosure;

[0020] FIG. 6 is a top view of the secondary face material/polishable layer of FIG. 5;

[0021] FIG. 7 is a side view of the polishable layer of FIG. 5, before the layer is combined with the green-state structure illustrated in FIGS. 4 and 5;

[0022] FIG. 8 is a top view of the multi-component device of FIG. 3;

[0023] FIG. 9 is a bottom view of the multi-component device of FIGS. 3 and 8;

[0024] FIG. 10 is a cross-sectional view of examples of parts removed from the multi-component device of FIGS. 3, 8 and 9, taken along line 10-10 of FIG. 8; and

[0025] FIG. 11 is a flow chart for an example of a method of making the parts illustrated in FIG. 10.

[0026] Throughout the drawings, like elements are designated by like reference numerals and other characters. The drawings show non-limiting examples for purposes of illustration and explanation of the present disclosure, and are not drawn to scale.

DETAILED DESCRIPTION

[0027] Referring now to the drawings, FIG. 3 shows an example of a multi-component, combined structure 40 constructed in accordance with the present disclosure. The multi-component structure 40 includes a substrate 42 (an example of a first component) and a polishable layer 44 (an example of a second component). The substrate 42 may be a layer (as illustrated in FIG. 3) or may have some other, suitable shape.

[0028] If desired, the multi-component structure 40 may have more than two components. The periphery 48 of the polishable layer 44 is not necessarily coextensive with the periphery 50 of the substrate 42. The present disclosure should not be limited to features shown in the drawings except to the extent such features are recited in the appended claims.

[0029] The substrate 42 is formed of the RB-SiC material 26 illustrated in FIG. 2. That is, the substrate 42 (FIG. 3)

includes a fully dense, three-component microstructure including original silicon-carbide particles 22 (FIG. 2), reaction-bonded silicon carbide 28, and residual (unreacted) elemental silicon 30.

[0030] The polishable layer 44 is formed of a suitable polishable material and does not include RB-SiC. The polishable layer 44 may include chemical-vapor-deposited silicon-carbide (CVD-SiC), single-crystal silicon-carbide (Xtal-SiC), or another suitable material or materials. The material of the polishable layer 44 may be selected so that polishable layer 44 has a good coefficient of thermal expansion (CTE) match to the substrate 42 across a wide range of temperatures. According to one aspect of the present disclosure, the difference in CTE between the substrate 42 and the polishable layer 44 is preferably less than 2.0 ppm/° C., and even more preferably less than 0.2 ppm/° C. As used herein, the phrase “a wide range of temperatures” means “a range of temperatures between ambient and the melting point of silicon.” The polishable layer 44 has a polishable surface 52. If desired, the surface 52 is polished to an extremely high dimensional accuracy and an extremely low roughness.

[0031] The polishable layer 44 may be formed in a chemical-vapor-deposition process, a single-crystal formation process, or another suitable process. According to one aspect of the present disclosure, the material of the polishable layer 44 has only one phase. In contrast to the substrate 42, the polishable layer 44 is not formed of a multi-phase material. As a result, it is easier to accurately polish the polishable surface 52 than it would be to accurately polish a surface of the substrate 42. If desired, the polishable layer 44 is formed and subjected to a suitable polishing process independently of (before or after) the infiltration process by which the substrate 42 is formed.

[0032] A green-state structure 60 for use in forming the substrate 42 is illustrated in FIGS. 4 and 5. The green-state structure 60 includes the same porous interconnected material as that of the porous mass 20 shown in FIG. 1. That is, the green-state structure 60 (FIGS. 4 and 5) includes interconnected silicon-carbide particles 22 (FIG. 1) and carbon 24.

[0033] The interconnected material of the green-state structure 60 (FIGS. 4 and 5) may be sufficiently cohesive and strong to permit movement and handling without degrading edges and surfaces of the green-state structure 60. The cohesiveness and strength of the interconnected green-state structure 60 may be established during the casting of the silicon-carbide particles 22 (FIG. 1) and carbon 24.

[0034] In operation, an unpolished surface 54 of the polishable layer 44 and a first surface 62 (also unpolished) of the green-state structure 60 are brought into contact with each other to form the preform 64 illustrated in FIG. 5. If desired, the green-state structure 60 may be cast onto the second surface 54 of the polishable layer 44. Then, molten elemental silicon (not illustrated in FIGS. 4 and 5) infiltrates into the green-state structure 60, within a vacuum or inert atmosphere, to produce the substrate 42. The infiltration process is essentially the same as the one used to produce the RB-SiC material illustrated in FIG. 2.

[0035] According to the present disclosure, however, the infiltration process which completes the formation of the substrate 42 may be continued such that some of the molten silicon exudes outwardly through the contacting surface 62, fills interstices between the contacting surface 62 and the

unpolished surface **54** of the polishable layer **44**, comes into contact with the unpolished surface **54**, and forms an interface **56** by reacting with silicon carbide in the polishable layer **44** to thereby securely bond the substrate **42** and the polishable layer **44** together.

[0036] After the infiltration process is completed, that is, after the combined structure **40** (FIG. 3) is formed, individual parts **80** (FIG. 10) may be extracted from the combined structure **40** using wire electrical-discharge-machining (EDM), spark machining, spark eroding, die sinking, wire burning, wire erosion, or another suitable process. The parts **80** may be employed as optical mirrors, beam dumps, or other optical devices. Surfaces **81** of the extracted parts **80** are formed of the polishable layer **44**. The surfaces **81** face away from the substrate **42** and do not contain any RB-SiC. The surfaces **81** may be polished after the substrate **42** is bonded to the polishable layer **44**.

[0037] The peripheries **82** of the extracted parts **80**, where the combined structure **40** is machined to extract the parts **80**, are designated in FIG. 8 by dotted lines. If desired, the extracted parts **80** may have peripheries and configurations different from the ones illustrated in FIGS. 8 and 10. As noted above, the present disclosure should not be limited to features shown in the drawings except to the extent such features are recited in the appended claims.

[0038] The processes described herein may be used to overcome or alleviate difficulties associated with creating an optical quality surface on an RB-SiC material. Such difficulties may be created, at least in part, by the multi-phase nature of RB-SiC. Different Young's moduli, yield strengths, hardnesses, and chemical reactivities of the different phases may make it difficult or impossible to polish RB-SiC to a desired, high degree of smoothness. In contrast to RB-SiC, chemical-vapor-deposited silicon-carbide (CVD-SiC) and single-crystal silicon-carbide (Xtal-SiC) can be polished with extremely high dimensional accuracy and extremely low roughness.

[0039] CVD-SiC, Xtal-SiC, and certain other single-phase, polishable ceramic materials, can be grown only in a limited number of shapes, which makes it difficult or impossible to use such materials in certain optical devices. RB-SiC, in contrast, can be configured in almost any shape through green-state machining, net-shape preform casting of the ceramic, or another suitable process, followed by an infiltration process. By combining a solid piece of single-phase, easily polishable material, such as CVD-SiC or Xtal-SiC, with a body of RB-SiC, one or more extremely high-quality mirrors **81** with independently shaped support structures can be produced.

[0040] Moreover, it may difficult or impossible to form a high-quality mirror, with properties equivalent to the polishable layer **44** of the present disclosure, by depositing a CVD-SiC coating on a RB-SiC substrate. Such a directly deposited coating may be too thin, irregular, or internally stressed to be acceptable. Moreover, additional operational time may be required to level the directly deposited coating, and production yield may be adversely impacted if the irregularity of the coating were to cause the polishing operation to break through the coated surface to the underlying RB-SiC material. Additionally, a direct-deposition process may result in the formation of unwanted deposits on one or more non-optical surfaces.

[0041] The present disclosure overcomes many or all of these potential challenges by bonding a bulk piece of a

material other than RB-SiC, such as CVD-SiC or Xtal-SiC, to an RB-SiC body **42** during an RB-SiC infiltration process. By casting a green-body ceramic **60** on a piece of CVD-SiC or Xtal-SiC, the crystalline SiC in the polishable layer **44** can bond to the ceramic RB-SiC during the infiltration process. If desired, the processes described above may be combined with RB-SiC to RB-SiC bonding methods to create more complex geometries like internal cooling channels (not illustrated).

[0042] If desired, the present disclosure may be implemented in connection with other ceramic substrates that are compatible with a reaction-bonding infiltration process. The RB-SiC substrate **42** illustrated in FIG. 3 is a non-limiting example of such ceramic substrates. The polishable surface **44** may be formed of any suitable material, other than the multi-phase material of the substrate, including, but not limited to, aluminum nitride (AlN), diamond, or a single-crystal silicon-carbide of any suitable allotrope or crystallographic orientation.

[0043] A method of producing the parts **80** illustrated in FIG. 10 is illustrated in FIG. 11. The method includes a step **100** of providing a green-state structure **60** and a polishable layer **44**, and a subsequent step **102** of infiltrating elemental silicon into and through the green-state structure **60**. In the illustrated method, the green-state structure **60** is preferably a porous mass of interconnected silicon-carbide plus carbon and is configured to receive molten elemental silicon by capillary action during the infiltrating step **102**.

[0044] As a result of the infiltrating step **102**, a substrate of fully dense, two-phase ceramic material is formed in place of the green-state structure, and an interface is formed by a reaction between the silicon and the material of the polishable layer, which interface bonds the substrate and the polishable layer together.

[0045] The processes described herein may be employed to produce a thick, uniform, face sheet of silicon carbide (or another suitable, polishable material) without any chemical-vapor-deposition occurring outside of optical area(s). Additionally, the processes described herein may have a relatively small number of process steps, and may be performed in an efficient and uncomplicated manner.

What is claimed as new and desired to be protected by Letters Patent of the United States is:

1. A method of making a multi-component device, comprising:
 - providing a green-state structure, wherein the green-state structure comprises a porous mass of interconnected silicon carbide and carbon;
 - providing a polishable layer;
 - contacting the polishable layer to the green-state structure;
 - after the contacting step, infiltrating silicon into and through the green-state structure to form a formed substrate having a multi-phase ceramic material from the green-state structure; and
 - bonding the substrate and the polishable layer together, wherein the polishable layer and the formed ceramic substrate have well-matched coefficients of thermal expansion (CTEs).
2. The method of claim 1, further comprising filling one or more interstices between the polishable layer and the green-state structure with the silicon.
3. The method of claim 2, further comprising contacting the silicon with the polishable layer.

4. The method of claim 1, wherein the ceramic material of the substrate includes reaction-bonded silicon-carbide (RB-SiC).

5. The method of claim 1 wherein the polishable layer comprises chemical-vapor-deposited silicon-carbide (CVD-SiC).

6. The method of claim 1 wherein the polishable layer comprises single-crystal silicon-carbide (Xtal-SiC).

7. The method of claim 1, wherein the polishable layer does not comprise reaction-bonded silicon-carbide (RB-SiC).

8. The method of claim 1, wherein the multi-component device comprises an optical device, and wherein the method further comprises extracting the optical device from the bonded substrate and polishable layer.

9. The method of claim 8, wherein the formed substrate comprises reaction-bonded silicon-carbide (RB-SiC).

10. The method of claim 8 wherein the polishable layer comprises chemical-vapor-deposited silicon carbide (CVD-SiC).

11. The method of claim 8, wherein the polishable layer comprises single-crystal silicon-carbide (Xtal-SiC).

12. The method of claim 8, wherein the step of extracting the optical device from the bonded substrate and polishable layer includes wire electrical-discharge-machining (EDM).

13. The method of claim 1, wherein a difference in CTE between the polishable layer and the formed ceramic substrate is less than 2.0 ppm/° C.

14. The method of claim 1, wherein a difference in CTE between the polishable layer and the formed ceramic substrate is less than 0.2 ppm/° C./.

15. The method of claim 1, wherein the polishable layer comprises at least one of aluminum nitride (AlN), diamond, and a single crystal silicon carbide of any suitable allotrope or crystallographic orientation.

16. The method of claim 1, wherein the multi-phase ceramic material comprises silicon carbide particles, reaction-bonded silicon carbide, and residual elemental silicon.

17. The method of claim 1, wherein the polishable layer is a single-phase polishable layer.

18. The method of claim 1, wherein molten silicon is infiltrated through the green-state structure through capillary action.

19. The method of claim 1, further comprising polishing the polishable layer.

20. The method of claim 19, wherein polishing the polishable layer occurs prior to the bonding step.

21. The method of claim 19, wherein polishing the polishable layer occurs after the bonding step.

* * * * *